

Round 2 — Employee CSV Upload

Technical Design · Implemented

L&D admin uploads a CSV. The API validates the file and each row, compares against tenant data, and returns a **preview** of proposed changes. **Nothing is written to the database.**

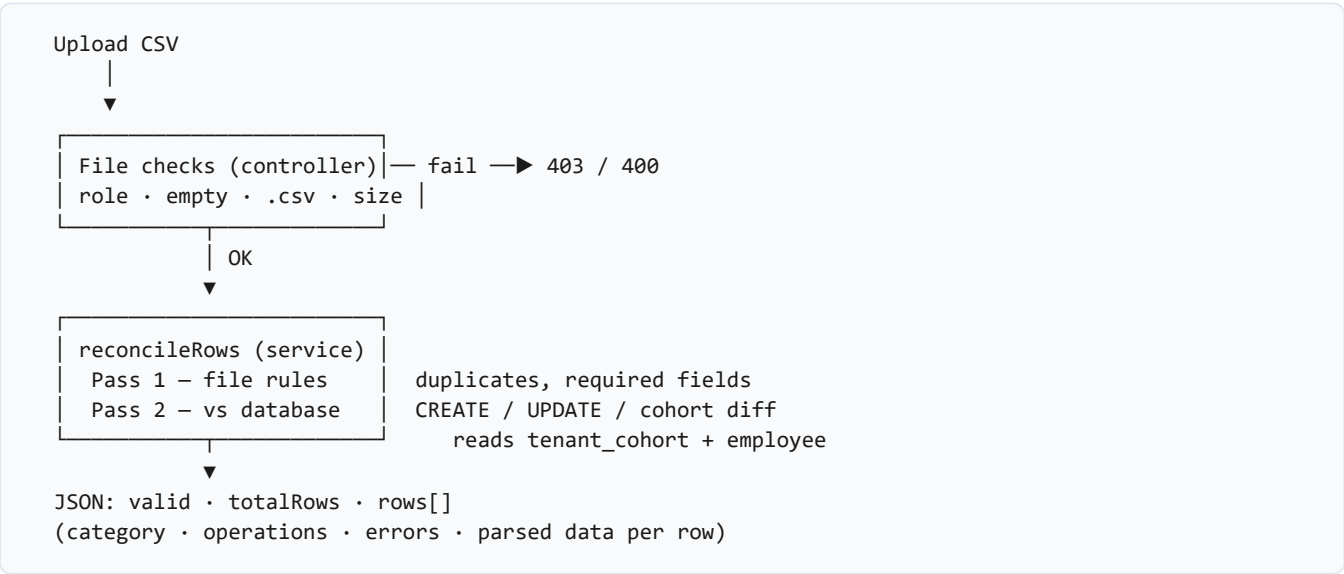
AT A GLANCE

Item	Detail
Endpoint	POST /v1/upload/emp-with-cohort-assignment
Headers	X-Tenant-Id · X-User-Role: LND_ADMIN · Idempotency-Key ¹
File	.csv , max 10 MB, multipart field file
All rows valid	200 OK + full response
Some rows invalid	400 + full response (every row still listed)
File rejected	403 / 400 + { "error": "..."}

¹ Header required; not stored or replayed in current code.

CSV columns: employee_id , email , name , optional role , optional cohort_names (semicolon-separated display names), optional start_date (parsed, not used in reconcile).

FLOW



COMPONENTS

Layer	Class	Role
Security	TenantAccessInterceptor	Requires tenant header
Controller	FileUploadController	File validation, orchestration
Service	FileUploadService	parseCSV() + reconcileRows()
Data	Cohort + Employee repos	Read-only lookups
Types	DTOs, enums	Response shape

Core: reconcileRows() — 18 unit tests + 7 integration tests.

DATABASE (READ PATH)

Table	Change	Used for
tenant_cohort	New	Map cohort display names from CSV
employee	+ name, email_id, role	Compare / preview create-update
employee_cohorts	Unchanged	Membership diff

Two reads per upload: all tenant cohorts + employees whose ids appear in the CSV.

VALIDATION

Whole file fails — wrong role (403); empty, bad extension, too large, parse error (400).

Row error	Meaning
REQUIRED_FIELD_MISSING	Missing id, email, or name
DUPLICATE_EMPLOYEE	Same employee id twice in file
DUPLICATE_EMAIL	Same email twice in file
UNKNOWN_COHORT	Cohort name not in tenant catalog
ORPHAN_COHORT_MEMBERSHIP	DB cohort id missing from catalog
EMAIL_CONFLICT	Email change blocked for existing employee

RECONCILE OUTCOMES

Situation	Category	Operations
New employee	CREATE	CREATE_EMPLOYEE (+ cohort add if listed)
No changes	SKIP_OP	NO_OP
Name / role change	UPDATE	UPDATE_EMPLOYEE
Cohort change	UPDATE	ADD / REMOVE_COHORT_MEMBERSHIP
Bad row	CORRECTION	NO_OP + errors

- CSV cohort list = **desired membership** (replace mode).
- Email change on existing employee = **error**.
- `valid: true` only when **every** row is valid.

DESIGN QUESTIONS

Topic	Answer (today)
Outcome	Preview JSON only — no DB writes
Processing	Synchronous; parse → 2 DB reads → in-memory reconcile
Partial failure	Row-level; one bad row does not hide others
Repeat upload	Idempotency header required; not replayed yet
Scale (~1.5k rows)	Load employees by CSV ids; reconcile in memory
Cohort names	Exact match to tenant catalog
Cohort list	Replace mode — CSV is source of truth
Email	No change via CSV; globally unique in DB

OUT OF SCOPE & TRADE-OFFS

- Not implemented: DB persistence, idempotency replay, mapper logic, `start_date` in reconcile.
- Trade-offs: extension-only `.csv` check · 10 MB in code · no email format validator · read-only.